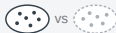


Allele-specific prediction is a nested solvability problem

A model can be judged only after the measurement detects a response and reproducibly identifies the allele.

1 DETECTION

variant vs wild type



2 IDENTIFICATION

variant vs siblings



3 PREDICTION

molecular features $\rightarrow \hat{\delta}_v$

correct held-out allele response?

Identification pass = allele-benchmarkable ground truth

Every inner claim requires every outer layer.

Verdict: where is the binding limit?

Detection or identification fails

MEASUREMENT-LIMITED

Ground truth cannot reward an allele model.
Do not rank models at allele resolution.

Identification passes; model < ceiling

MODEL-LIMITED

A meaningful computational gap.

Model reaches the measurement ceiling

RESOLVED AT CURRENT CEILING

Prediction is supported at this resolution.

! Direction recovered (Pearson- δ) $\neq$ allele identified or predicted

Held-out task

observed cells define δ_v and resolution

• model sees only θ /ESM and predicts $\hat{\delta}_v$